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Paper Title : Survey On Real Time Helmet Violation Detection Using Deep Learning

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AGENDA

- Abstract
- Introduction
- Literature Review
- Methodology
- Analysis
- Results & Discussion
- Conclusion

ABSTRACT

- Artificial Intelligence (AI) is widely used to solve real-life problems by enabling machines to perform intelligent tasks automatically. Deep learning techniques in computer vision help systems analyze images and videos for applications such as traffic monitoring and road safety.
- The existing system proposed an AI-based framework with a lightweight classifier to detect helmetless motorcycle riders. The system first identifies motorbike riders from traffic videos and then checks whether they are wearing helmets. Although the model provides good accuracy with low computational cost, it mainly focuses on helmet detection and was tested under limited conditions.
- The proposed system improves this approach by using DetectNet and YOLOv8 algorithms to detect multiple traffic violations such as helmetless riding, triple riding, and mobile phone usage. The system uses a more diverse dataset and can work under different environmental conditions. It also generates automatic alerts to authorized authorities, helping improve real-time traffic monitoring and road safety.

- **Keywords:** Artificial Intelligence, Deep Learning, Helmet Detection, Traffic Safety, Computer Vision.

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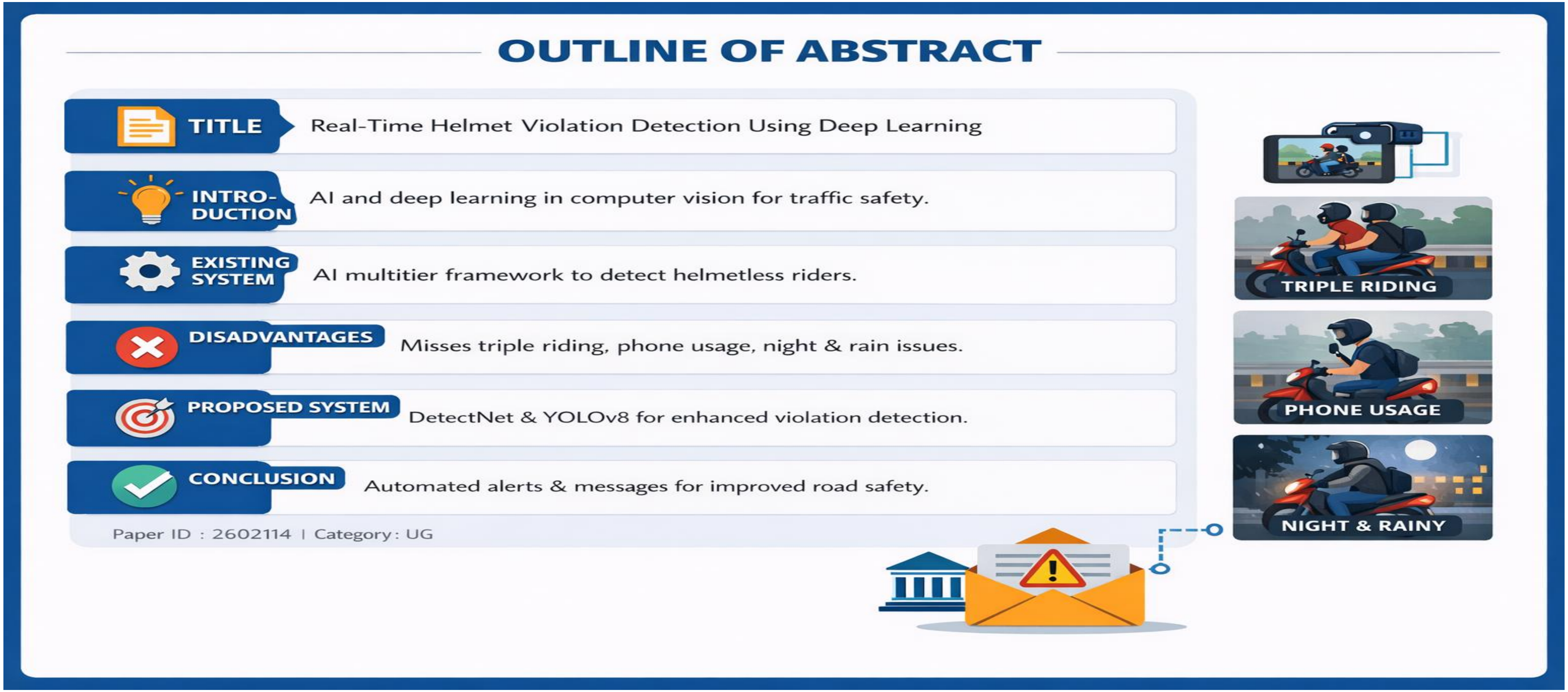
OUTLINE OF ABSTRACT:



INTRODUCTION

- Road accidents involving two-wheeler riders are often caused by not wearing helmets. Manual checking of helmet violations is inefficient and time-consuming. This project introduces a **real-time helmet detection system** using **deep learning** to automatically identify riders without helmets from live video feeds. The system detects motorcycles and classifies helmet usage accurately, helping traffic authorities enforce rules and **improve road safety efficiently**

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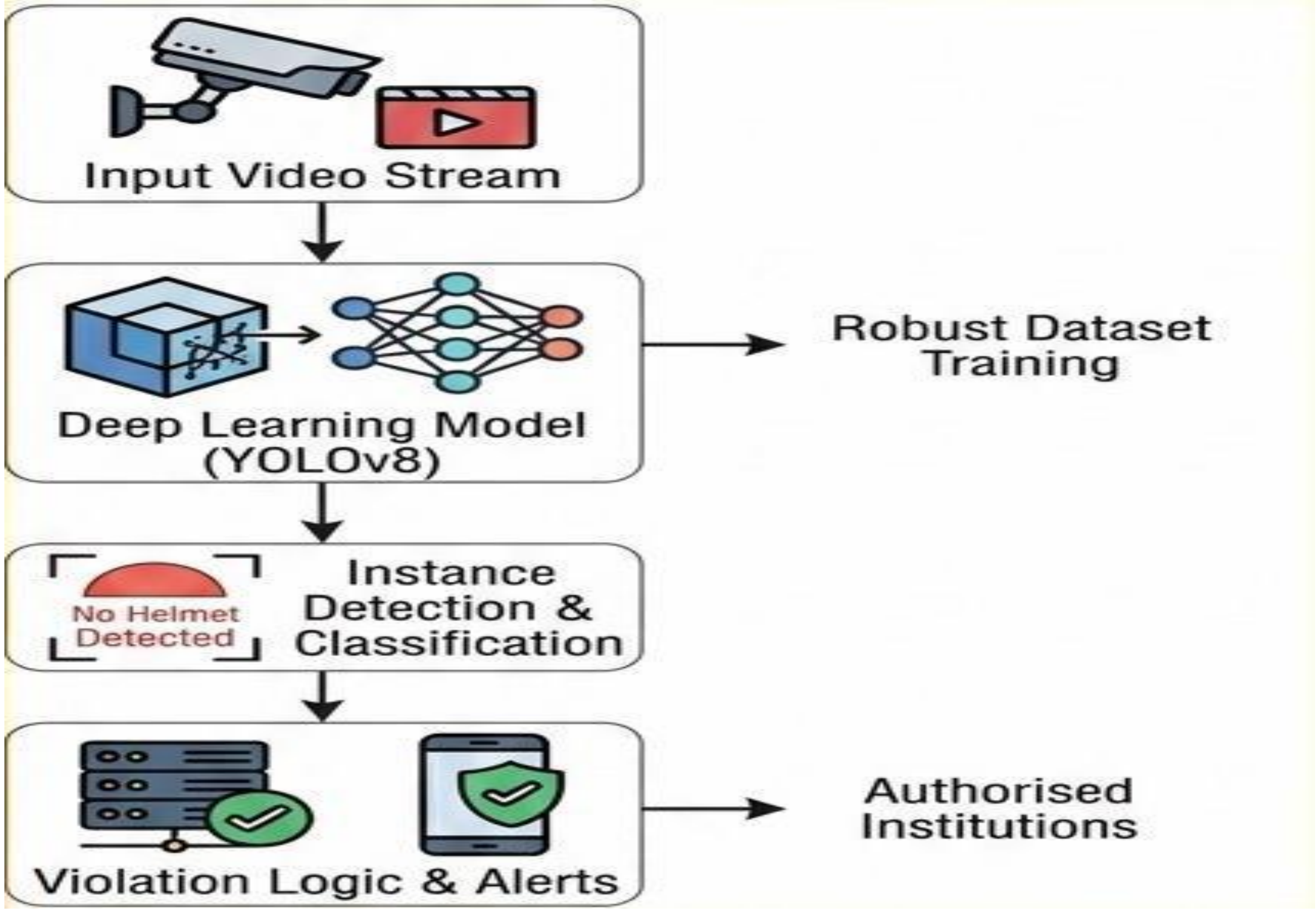
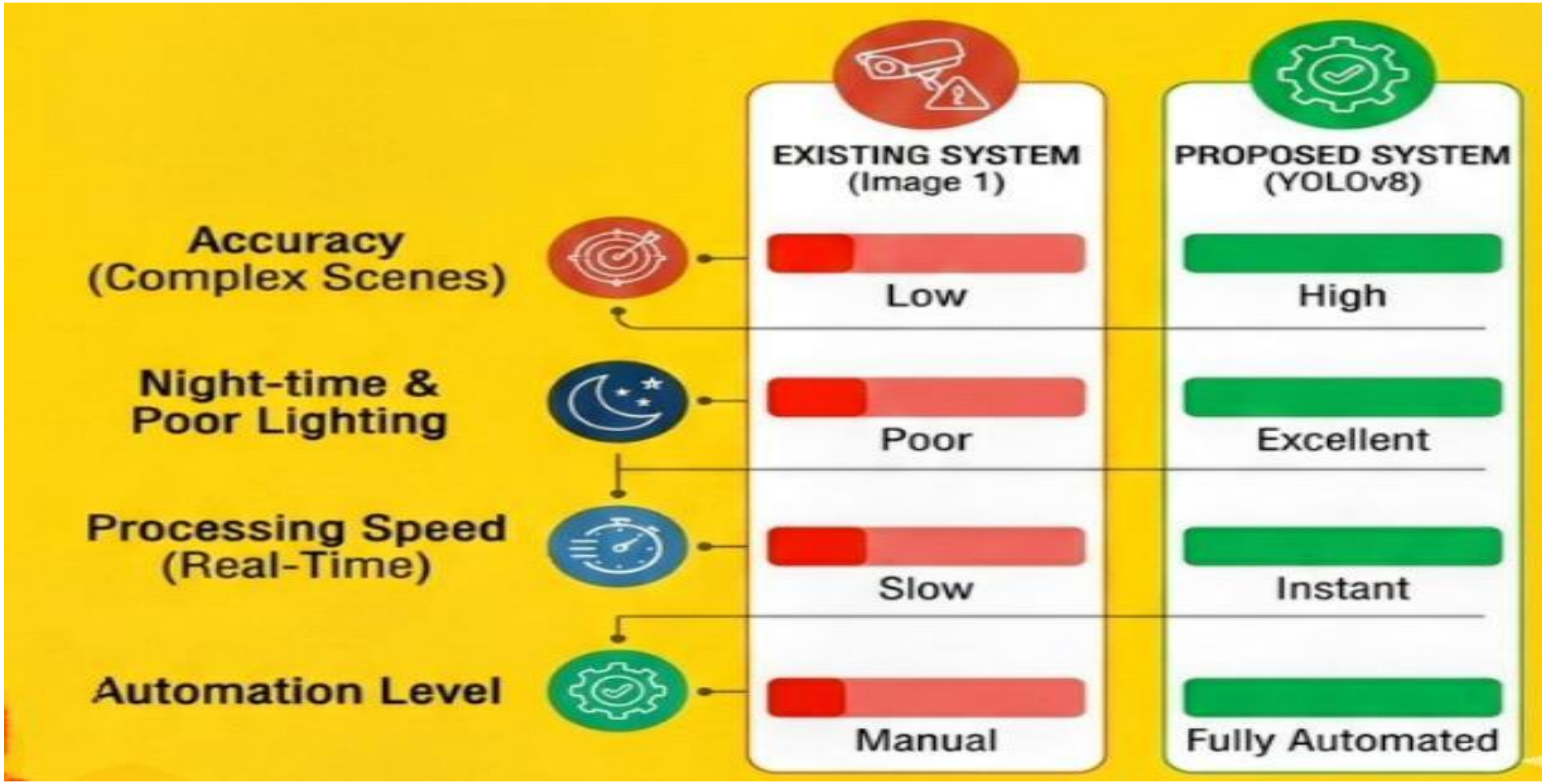
LITERATURE REVIEW

- Several research works have been carried out in the area of helmet violation detection using deep learning techniques. An AI-based helmet violation detection system was proposed using deep learning models for real-time traffic monitoring. The system used optimized neural network models to improve detection accuracy while reducing computational complexity. The results showed that deep learning techniques can provide reliable and efficient helmet detection for traffic safety applications.
- Another study proposed a two-stage helmet violation detection method using DetectNet_v2 and YOLOv8 algorithms. In this approach, DetectNet_v2 was used to detect motorcycle riders, while YOLOv8 was used to classify helmet usage and detect license plates. The system was trained using a custom dataset and achieved high accuracy in real-time traffic conditions. This study demonstrated that combining multiple deep learning models improves detection accuracy and system performance.

METHODOLOGY



ANALYSIS


EXISTING SYSTEM
(Image 1)
PROPOSED SYSTEM
(YOLOv8)
Accuracy
(Complex Scenes)
Low
High
Night-time &
Poor Lighting
Poor
Excellent
Processing Speed
(Real-Time)
Slow
Instant
Automation Level
Manual
Fully Automated

RESULTS & DISCUSSION

- YOLOv8 model successfully detects helmet and no-helmet riders in real time. High accuracy achieved under normal lighting conditions.
- Faster detection speed compared to traditional image processing methods. Real-time alert system works effectively for violation notification.
- Performance may reduce slightly in night or rainy conditions.
- Accuracy depends on dataset quality and training diversity.
- System reduces manual monitoring and supports smart traffic enforcement.

CONCLUSION

- The project *Real Time Helmet Violation Detection Using Deep Learning* successfully uses AI and YOLOv8 to detect helmet violations in real time.
- The system provides accurate and fast detection compared to manual monitoring methods.
- It automatically generates alerts, reducing human effort and improving traffic rule enforcement.
- Overall, the proposed system enhances road safety and supports smart city surveillance applications.

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